generation) air-to-ground missiles, Astra air-to-air missiles, anti-tank missiles, Rudram air-to-ground anti-radiation missiles, laser-guided bombs and precision munitions.

Dr Ashish Kumar Ghosh, known as the 'AMCA man' since he was its project director since inception, tells INDIA TODAY that designers at the ADA have been waiting for the government's nod for years. Underlining the project's importance for national security, he says that after years of hard work, India finally has developed its own stealth technology. Now retired, Ghosh also points out that not all fifth generation jets have every 'stealth' feature. "But India's AMCA is going to be a real stealth jet," he says. The AMCA, emphasises Ghosh, is not a copy of any existing platform. It also must be mentioned that the AMCA's design and the development of its various technologies happened in tandem.

Dr Girish Deodhare, former head of the ADA, says he is delighted with the government's go-ahead. "We have been working on the AMCA for over a decade. Government sanction came at the right time, as most of the technologies are matured."

Another senior scientist involved in the project points to the all-round effort taken to develop the cutting-edge stealth technologies. "For LCA (Light Combat Aircraft) Tejas, we had some consultancies. But when fifth gen technologies came up, these consultants would clam up. So, all the technologies have been developed indigenously, with many trials and errors. It has been a truly national effort," he says. Agencies involved included DRDO labs, academic institutions, CSIR (the Council for Scientific and Industrial Research) labs, PSUs and private agencies.

THE LONG STRUGGLE

In 2005, after the US launched the F22 Raptor, India formed a study group to consider fifth generation fighter jet capabilities. Subsequently, in 2009, a budget of Rs 90 crore was given to the ADA to carry out a feasibility study for a stealth fighter jet. Four years later in 2013, AMCA developers worked out the first feasible configuration, which was accepted by the IAF.

But, in between in 2010, enthused with the success of the India-Russia BrahMos cruise missile project, both nations agreed to develop an FGFA in a joint venture. A preliminary design contract was inked, with the agreement that Russia would procure 250 and India 144 aircraft at a cost of around $30 billion (Rs 2.46 lakh crore) by 2022.



"STEALTH AIRCRAFT HAVE TO BEAT THE ENEMY'S AIR DEFENCE SENSORS BEFORE HITTING TARGETS. THEY HAVE NON-STEALTH ROLES TOO"

AIR MARSHAL S.B.S. SINHA (RETD)
Former deputy chief, IAF

After working jointly for years, India pulled out of the project in 2018 over differences with Russia on issues like the transfer of technology and design specifications of the aircraft. That same year, the government approved an additional Rs 447 crore for completion of the design and development of stealth technologies for India's future fifth gen fighter.

Over the past decade, the absence of a roadmap for jet engines slowed down India's fighter jet programme, including the AMCA project, as New Delhi negotiated with Washington for the joint production of GE 414 engines in India. With the US agreeing to share jet engine technology with India in 2023, the first 40 AMCAs are expected to fly on these GE engines. The advanced AMCA Mark II version will need a more powerful 110kN high-thrust engine for better supercruise abilities, which will be co-developed in India.

Since a fifth generation stealth fighter is an extremely high-cost platform, no country with the capability has it in large numbers. Likewise, the IAF has expressed a need for six squadrons of the AMCA (including four of the Mk-II version). The first five prototype jets will cost nearly Rs 900 crore each, but the cost will reduce at the production stage.

All sensors, avionics and flight control systems of the AMCA are to be indigenous—over 70 per cent of the aircraft—except for the GE-414 engine and the ejection seat. "Since AMCA is an indigenous fifth generation aircraft, it is less costly than similar aircraft available outside," Ajay Bhatt, the minister of state for defence, stated in Parliament in 2022.

By the IAF's reckoning, the fifth generation fighter would form an important element of future air combat. Air Marshal S.B.S. Sinha (retd), former deputy chief of the IAF, points out that centres of gravity (points of strength of a military) of every country are heavily defended by a mix of long-range sophisticated missile systems. "Therefore, you require stealth aircraft to beat enemy's AD (air defence) sensors to neutralise high-value targets, while other fighters continue routine missions. Stealth aircraft can also be used in non-stealth roles, wherein it can carry heavy underwing and underbelly weapons like normal fighters," he adds.

Ghosh says that now that the government has allocated the budget, the prototypes must be made at the earliest. He stresses that the capability of manufacturing the AMCA also needs to be developed speedily once a final version has been certified. "Only then can we get the best of its next generation features," he says. ■